

Risk Measures Comparison (\$1M Portfolio)  
VaR and Expected Shortfall under Different Distributions

| Level | Normal VaR | Normal ES | Student-t VaR | Student-t ES | Empirical VaR | Empirical ES |
|-------|------------|-----------|---------------|--------------|---------------|--------------|
| 95%   | \$19,853   | \$24,957  | \$16,834      | \$29,902     | \$18,659      | \$29,847     |
| 99%   | \$28,178   | \$32,317  | \$35,036      | \$58,092     | \$34,839      | \$50,602     |
| 99.9% | \$37,508   | \$40,890  | \$87,603      | \$142,040    | \$73,903      | \$95,179     |

Data: S&P 500 log-returns, 2000–2025 (n=6,537)  
Student-t df=2.64 | ES values shown in red (ES ≥ VaR by construction)